

Light in Medicine – Short Essay Questions with Answers

1. 1. Explain how light is detected in the eye.
2. 2. Describe image formation in the normal eye.
3. 3. What is visual acuity?
4. 4. Define eye defects.
5. 5. List the common types of eye defects.
6. 6. Define myopia.
7. 7. Explain the causes of myopia.
8. 8. Describe the symptoms of myopia.
9. 9. How is myopia diagnosed?
10. 10. Explain the correction of myopia.
11. 11. Why are concave lenses used in myopia?
12. 12.
13. 13. What is degenerative myopia?
14. 14. Define hypermetropia.
15. 15. Explain the causes of hypermetropia.
16. 16. Describe the symptoms of hypermetropia.
17. 17. How is hypermetropia diagnosed?
18. 18. Explain the correction of hypermetropia.
19. 19. Why are convex lenses used in hypermetropia?
20. 20. Define presbyopia.
21. . Compare myopia and hypermetropia.
22. Compare correction lenses for myopia and hypermetropia.
23. Explain how LASIK and PRK correct vision.
24. Discuss the importance of accommodation.
25. Summarize major eye defects and their corrections.
26. Define visual acuity and list its main determinants.
27. What is the principle of Snellen's chart?
28. What does 6/6 vision mean?
29. Describe the procedure for using Snellen's chart.
30. What is the Landolt C chart and why is it important?
31. Explain the tumbling E chart and its clinical use.
32. What is the LogMAR chart?
33. Define slit lamp biomicroscope and its function.
34. Mention three conditions detected by slit lamp examination.
35. What is tonometry and why is it important?
36. Describe the principle of contact tonometry (Goldmann tonometer).
37. What is the normal range of intraocular pressure?
38. Explain non-contact (air puff) tonometry.
39. What is an automated tonometer?

40. Define retinoscope and its clinical use.
41. What is retinoscopy?
42. Why are pupil-dilating drops used in retinoscopy?
43. What is a phoropter and its purpose?
44. Describe the main steps of the phoropter test.
45. What is an autorefractor?
46. Define ophthalmoscope and its function.
47. Differentiate between direct and indirect ophthalmoscopy.
48. Why are dilating drops used before ophthalmoscopy?
49. Define keratometer and its function.
50. What parameters does keratometry measure?
51. State two clinical uses of keratometry.
52. What is corneal topography?
53. What do warm and cool colors indicate on corneal topography maps?
54. Mention three clinical applications of corneal topography.
55. Define endoscope.
56. List three main uses of endoscopy.
57. What is laparoscopy?
58. What factors determine the quality of endoscopic images?
59. Define cataract and explain its causes.
60. Describe the types of cataract.
61. Explain how cataract is diagnosed.
62. Describe the treatment and management of cataract.